

Aim Programs to input data through Command line argument.

Software Required:- Turbo C/C++

Theory:- Command-line arguments are given after the name of the program in Command line-shell of operating systems.

To pass command line arguments, we typically define `main()` with two arguments: first argument is the number of command line arguments & second is list of command-line arguments.

Syntax:-

```
#include <stdio.h>
int main (int argc, char *argv[])
{
    /* ..... */
}
```

In the above syntax;

- `Arg` (Argument Count) is `int` and stores number of command line argument passed by the user including the name of the program. So if we pass a value to a program, value of `argc` would be 2 (One for argument & one for program name).
- The value of `argc` should be non negative.
- `argv` (argument vector) is array of character pointers listening all the argument.
- If `argc` is greater than zero, the array elements from `argv[0]` to

Output :-

After executing the program, go to DOS shell from FILE option.

```
C:\Turbo C3\Bin>
C:\Turbo C3\Bin>ed..
C:\Turbo C3>cd source
C:\Turbo C3\Source>program.exe
please provide the arguments.
C:\Turbo C3\Source>program is one
the command line argument is one
C:\Turbo C3\Source>program.exe
one direction
more than 1 argument
```

- In `argv[argc-1]` will contain pointer to string.
- `argv[0]` is the name of program. After that till `argv[argc-1]` every element is command-line argument.

Code

```
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>

void main (int argc, char* argv[])
{
    clrscr();
    if (argc == 2)
    {
        printf("The command line argument is\n%s", argv[1]);
    }
    else if (argc > 2)
    {
        printf("More than 1 arguments");
    }
    else
    {
        printf("Please provide the arguments");
    }
    getch();
}
```


Output

After executing the program, go to Dos Shell from file option.

C:\Turbo C3\Bin>

C:\Turbo C3\Bin>cd.

C:\Turbo C3>cd source

C:\Turbo C3\Source>program.exe

PLEASE, PROVIDE THE ARGUMENTS

C:\Turbo C3\Source>program.exe 22

The sum of all command line program arguments is 22

C:\Turbo C3\Source>program.exe 22 21 24

The sum of all command line program arguments is 67

Code

```
#include <stdio.h>
#include <conio.h>
#include <stdlib.h>

void main (int argc, char *argv[])
{
    clrscr();
    int i, n;
    int sum = 0;
    if (argc == 1)
    {
        printf("PLEASE PROVIDE THE COMMAND LINE ARGUMENTS.");
    }
    for (i = 1; i < argc; i++)
    {
        n = atoi(argv[i]);
        sum += n;
    }
    printf("The sum of all command line arguments: %d", sum);
    getch();
}
```

Signature
13/12